Engine Model(s): PT6A

73-20-00

FUEL CONTROL UNIT - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03 which is 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC01-001	Fuel, Engine
PWC03-001	Oil, Engine
PWC03-002	Fluid, Calibrating
PWC05-061	Cloth, Abrasive Coated Crocus
PWC05-077	Oil, Preservative
PWC05-101	Cloth, Abrasive
PWC05-166	Solution, Chemical Treatment
PWC05-168	Compound, Polishing
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine
PWC11-038	Solvent, Cleaning

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Name	Remarks
FCU Shipping Stand	P/N 2529654

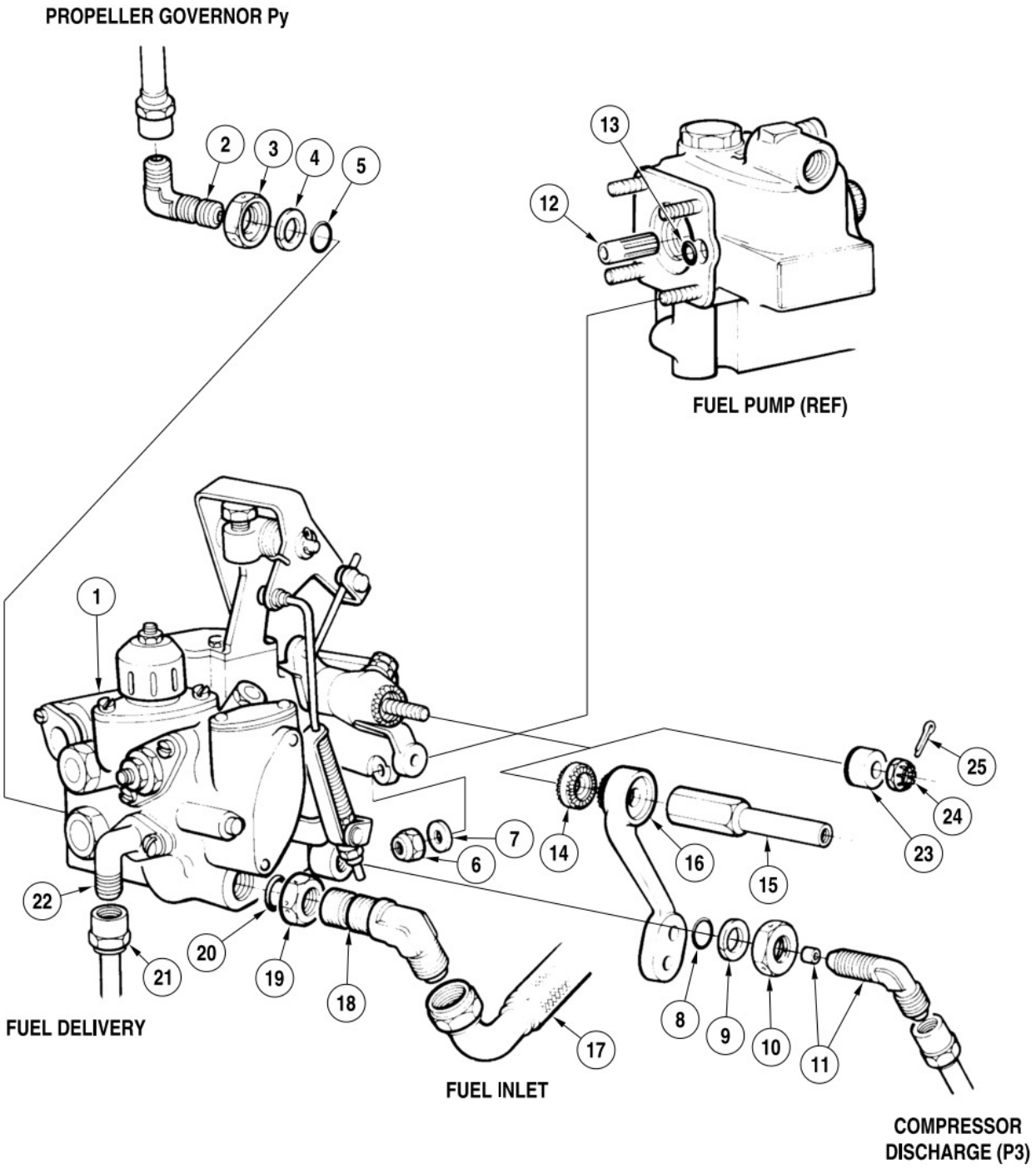
5. Removal/Installation

A. Removal of Fuel Control Unit (PT6A-21 Engines) (Ref. Fig. 201)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 201 Removal/Installation of Fuel Control Unit

(SHEET 1 OF 2)

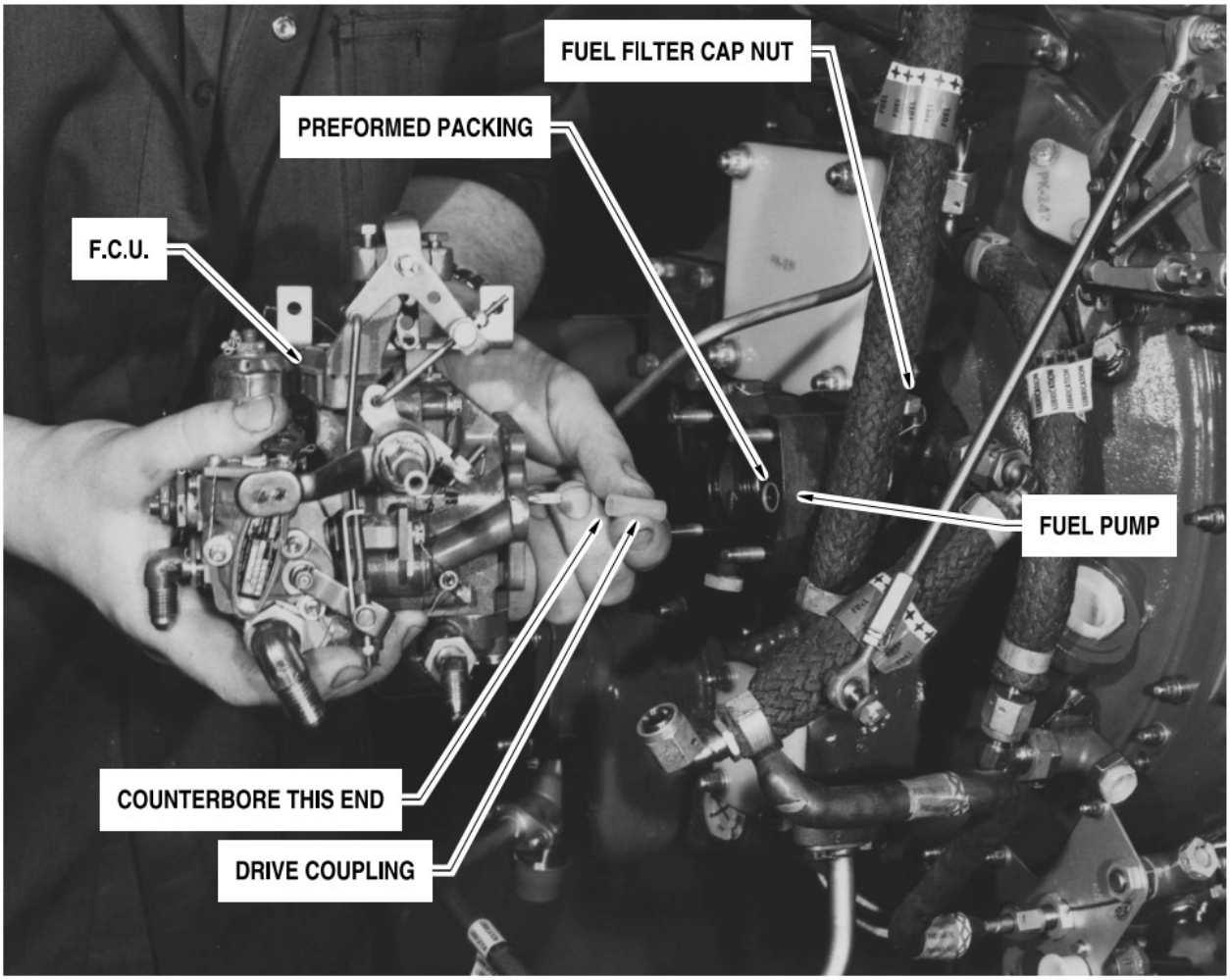


PT6A-21

c7915a

(SHEET 2 OF 2)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement



PT6A-21

c618a

1. Fuel Control Unit (FCU)
2. Elbow
3. Locknut
4. Back-up Ring
5. Preformed Packing
6. Self-locking Nut
7. Washer
8. Preformed Packing
9. Back-up Ring
10. Locknut
11. Elbow and Metering Plug
12. Coupling
13. Preformed Packing
14. Serrated Spacer
15. Arm Extension
16. Fuel Control Arm
17. Fuel Input Hose
18. Elbow
19. Locknut
20. Preformed Packing
21. Fuel Delivery Line
22. Fuel Delivery Elbow
23. Spacer
24. Castellated Nut
25. Cotterpin

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

NOTE: If the FCU is to stored off the engine, prepare the FCU for short or long term storage (Ref. Storage) after removal is complete.

(1) Disconnect fuel input hose from elbow fitting (18) on fuel control unit (1). Cap end fittings.

(2) Disconnect the fuel delivery line from the elbow fitting (22) on the FCU. Cap the end fittings.

NOTE: Make sure that both the ends of the delivery line are removed before the FCU removal from the engine.

(3) Disconnect coupling nuts of compressor delivery tube (P3) and propeller governor tube (Py) from respective elbow fittings (11 and 2) on FCU. Cap end fittings.

(4) Remove cotterpin, castellated nut, washer and bolt securing FCU reversing interconnect rod to fuel control unit arm (16).

(5) Remove four self-locking nuts (6) and washers (7) that secure fuel control unit (1) to fuel pump. Remove FCU from mounting studs.

(6) Remove fuel control-to-fuel pump coupling (12) and preformed packings (13).

(7) If fuel control unit is being replaced, remove fuel control arm extension (15), fuel control unit arm (16) and serrated spacer (14), and retain for installation on replacement unit. Install spacer (23), castellated nut (24) and retain with cotterpin (25) on removed unit.

NOTE: It is assumed shipping parts, items 23, 24 and 25 are available.

(8) Remove elbow fittings (2 and 11) and associated packings, retainers and jam nuts from respective ports on FCU, and retain for installation on replacement unit.

NOTE: Do not remove elbow fittings (18 and 22).

B. Installation of Fuel Control Unit (PT6A-21 Engines) (Ref. Fig. 201)

CAUTION: VARIOUS COMPONENTS ON THE FUEL CONTROL UNIT ARE LOCKWIRED AND SEALED. ENSURE ALL SUCH SEALS ARE INTACT PRIOR TO INSTALLATION AND ARE NOT TAMPERED WITH. UNITS WITH BROKEN SEALS MUST BE RETURNED TO OVERHAUL FOR RECALIBRATION.

NOTE: When a replacement FCU is being installed, check the length of storage and do reconditioning after storage (Ref. Storage).

(1) When a replacement fuel control unit (1) is being installed, depreserve, and assemble fittings (Ref. Para. A.) prior to installation:

NOTE: Depreservation is not required when original fuel control unit is being installed.

(a) Remove shipping caps from fuel inlet and outlet fittings. Ensure that air section of unit is sealed with shipping plugs and/or caps in Py and P3 ports.

(b) Drain as much residual preservation fluid as possible from fuel section of unit.

(c) Flush fuel section of unit with fuel (PWC01-001) that has been passed through a 10-micron (nominal) filter, until all preservation fluid has been removed. Ensure residual

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

flushing fuel is drained from unit.

(d) Remove shipping plug from P3 port and assemble nut (10), back-up ring (9), and preformed packing (8), on elbow and metering plug (11). Install elbow in P3 inlet port of unit. (Ref. 70-00-00.) Cap elbow to prevent ingress of foreign material until unit is ready for installation.

NOTE: After installation of elbow and metering plug (11), do not final torque locknut (10). Final torquing will be done at a later stage of installation.

(e) Remove shipping plug from Py port and assemble nut (3), back-up ring (4), and preformed packing (5) on elbow (2). Install elbow in Py port of unit (Ref. 70-00-00.) Cap elbow to prevent ingress of foreign material until unit is ready for installation.

CAUTION: IF THE FCU SPLINE AND THE PLASTIC DRIVE COUPLING (12) ARE NOT FULLY ENGAGED, IT COULD PUSH AND DISENGAGE THE FUEL PUMP DRIVE SHAFT WHEN MATED WITH THE FUEL PUMP ASSEMBLY.

(2) Install and push the plastic drive coupling (12) on the FCU shaft to make sure that the coupling is correctly engaged on the FCU shaft such that there is no more visible spline is found.

(3) Install preformed packing (13) in recess in mounting face to fuel pump; lubricate with engine oil (PWC03-001).

(4) Make sure that the fuel control unit driveshaft bearing retainer plate and surround area is clean. Remove all traces of blue grease or blue dye residue.

CAUTION: MAKE SURE THAT THE EACH STUD PROTRUDES A MINIMUM OF 0.054 IN. FROM EACH NUT.

CAUTION: CARE MUST BE TAKEN TO ALIGN THE SPLINES. DO NOT FORCE THE SPLINES TO ENGAGE THE FCU AND FUEL PUMP AS THIS COULD DISENGAGE THE PUMP SHAFT.

(5) Put the fuel control unit (1) on the studs of the fuel pump. Push in to engage the splines of the coupling (12) with the splines of the pump driveshaft. As the mounting faces meet, make sure that the preformed packing (13) is not displaced. Attach the fuel control unit (1) to the fuel pump with the washers (7) and the nuts (6). Torque the nuts 75 to 85 lbf.in.

(6) When a replacement fuel control unit is being installed, remove shipping components, item (23, 24 and 25) from FCU and retain. Install serrated spacer (14), together with FCU arm (16) on FCU and engage serrations with arm set at approximate position to that on original unit. Secure with FCU arm extension (15) and tighten fingertight.

(7) Connect FCU reversing interconnect rod and adjust as required (Ref. 76-10-00) . Align inner hole in FCU arm (16) with hole in rod end connector of FCU reversing linkage by rotating arm (16) and serrated spacer (14) around on serrations. Alignment is achieved when a bolt can be inserted freely through holes in arm and rod end connector. Refer to Rigging Instructions in relevant airframe manual.

(8) After adjustment, tighten extension (15), torque 25 to 35 lb.in., and lockwire.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(9) Secure FCU reversing interconnect rod to inner hole in FCU arm (16) with bolt, washer and castellated nut. Tighten nut and torque 12 to 18 lb.in., ensuring slots in nut are aligned with hole in bolt within torque limit. Install and lock cotterpin.

NOTE: Do not lockwire connections of air pressure lines. Lockwiring must be done after pressure test (Ref. 73-10-07, Adjustment/Test.)

(10) Connect coupling nuts of compressor delivery (P3) and propeller governor (Py) tubes to elbow fittings (11 and 2) respectively. Final torque locknuts (10 and 3) respectively 38 to 42 lb.in. Tighten coupling nuts, torque 90 to 100 lb.in.

(11) Connect coupling nut of fuel inlet hose (17) to elbow fitting (18) on FCU. Tighten nut, torque 270 to 300 lb.in., and lockwire.

CAUTION: IT MAY BE NECESSARY TO CHANGE THE ANGULAR POSITION OF FUEL CONTROL INLET ELBOW (18) TO ACHIEVE CORRECT ALIGNMENT OF HOSE (17), IN WHICH CASE INSTRUCTIONS IN STANDARD PRACTICES (REF. 70-00-00) MUST BE OBSERVED TO AVOID DAMAGE TO THE PREFORMED PACKING, WITH RESULTANT LEAKAGE.

(12) Connect coupling nut of fuel delivery tube (21) to elbow fitting (22) on FCU. Tighten nut, torque 90 to 100 lb.in., and lockwire.

(13) Do pressure leak test on FCU pneumatic system (Ref. 73-10-07).

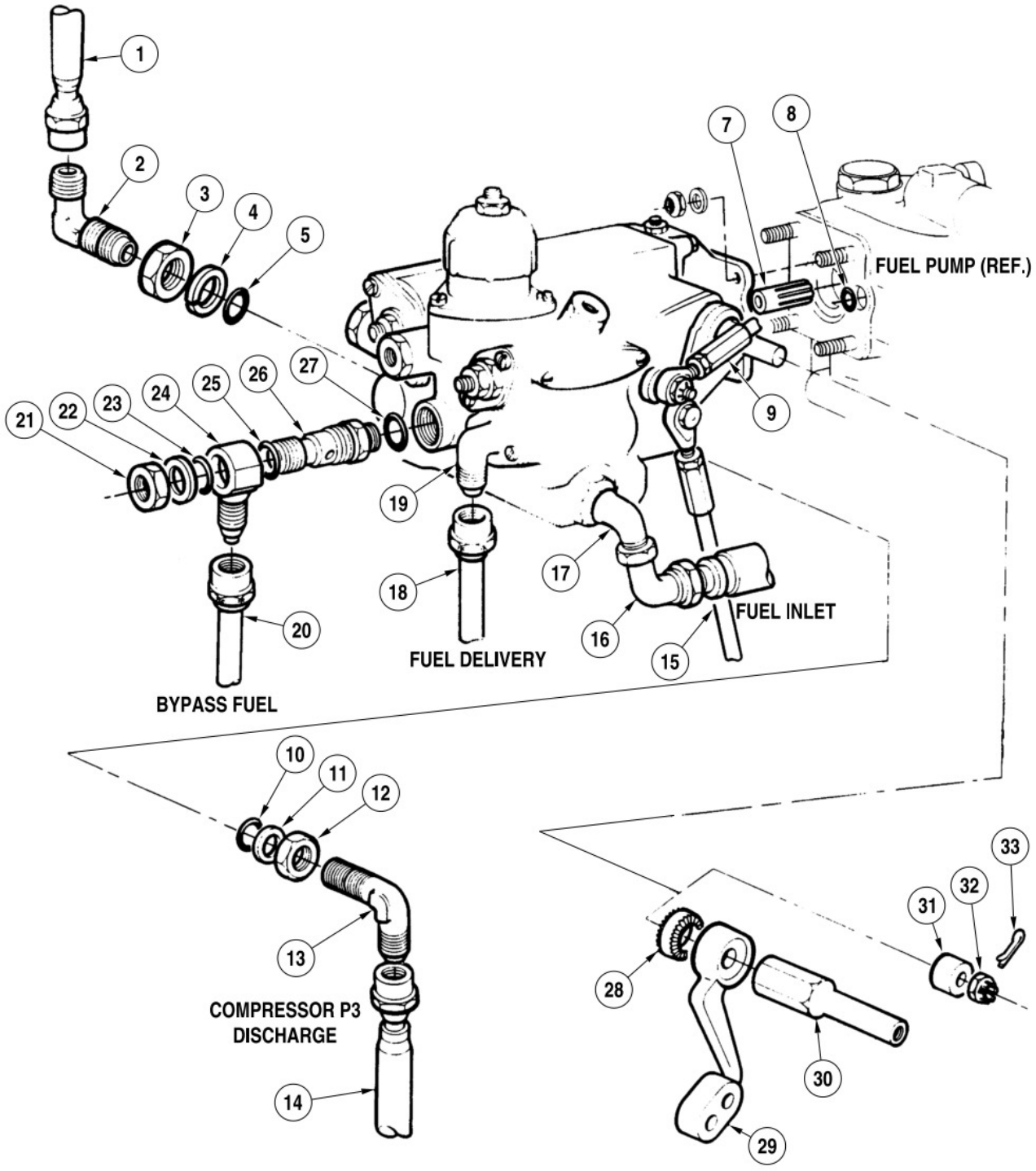
(14) Check FCU operation and for leakage at next engine test (Ref. Adjustment/Test).

C. Removal of Fuel Control Unit (PT6A-27 and PT6A-28 Engines) (Ref. Fig. 202)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 202 Removal/Installation of Fuel Control Unit

(SHEET 1 OF 2)

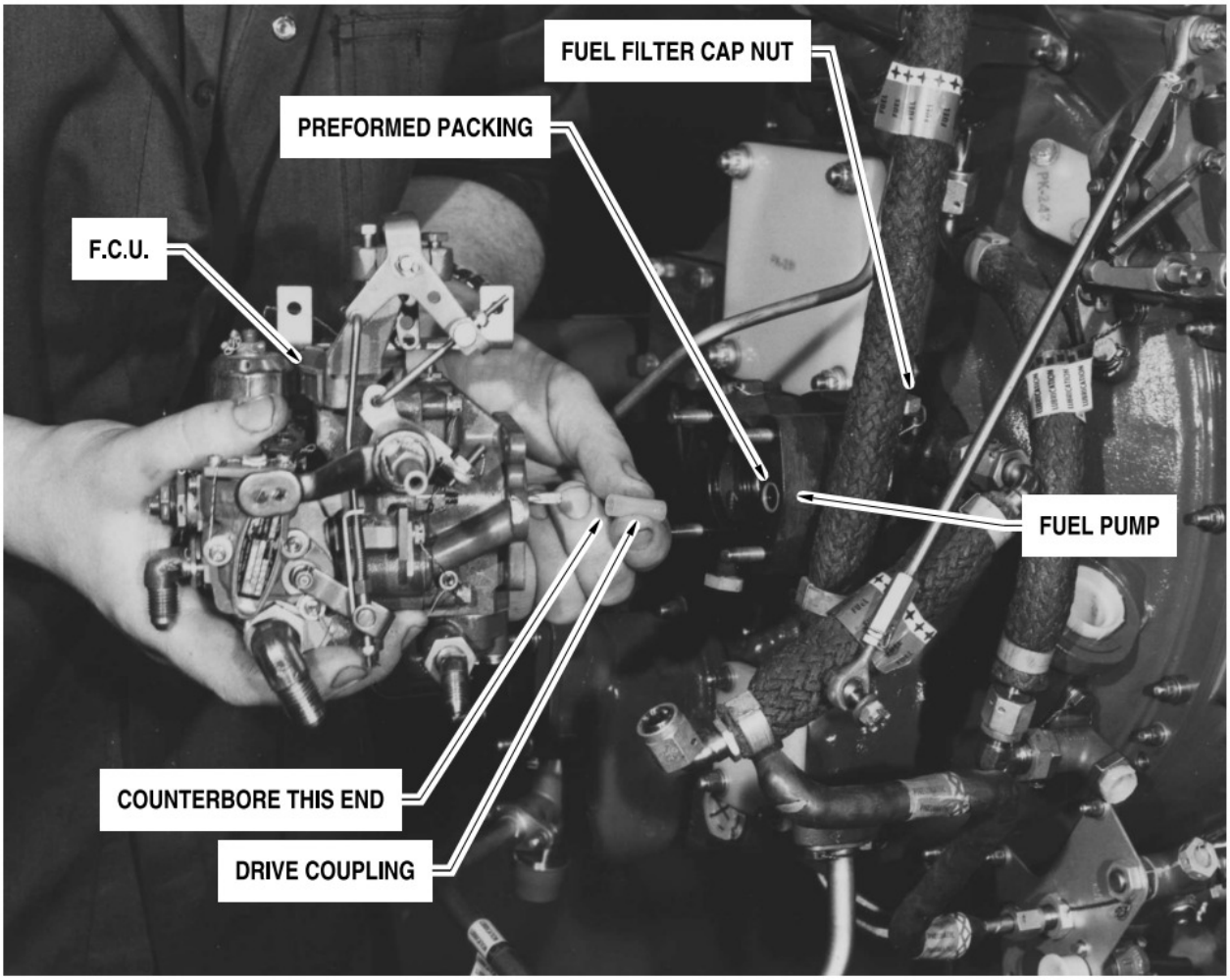


PT6A-27 and PT6A-28

c7914

(SHEET 2 OF 2)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement



PT6A-27 and PT6A-28

c618a

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Air Pressure Line (Py)
2. Elbow
3. Locknut
4. Back-up Ring
5. Preformed Packing
6. Fuel Control Unit
7. Coupling
8. Preformed Packing
9. Control Rod (to cambox)
10. Preformed Packing
11. Back-up Ring
12. Locknut
13. Elbow and Metering Plug (Unmettered on Pre-SB1123 engines)
14. Air Pressure Line (P3)
15. Control Rod (to start control unit)
16. Fuel Inlet Line
17. Elbow
18. Fuel Delivery Line
19. Fuel Delivery Elbow
20. Fuel Bypass Line
21. Locknut
22. Back-up Ring
23. Preformed Packing
24. Fluid Connector
25. Preformed Packing
26. Fluid Passage Bolt
27. Preformed Packing
28. Serrated Spacer
29. Fuel Control Arm
30. FCU Arm Extension
31. Spacer
32. Castellated Nut
33. Cotterpin

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

NOTE: If the FCU is to stored off the engine, prepare the FCU for short or long term storage (Ref. Storage) after removal is complete.

(1) Provide suitable drip pan for residual fuel spillage.

(2) Disconnect the following lines from fuel control unit (6):

(a) Fuel inlet line (16) from elbow (17).

(b) Fuel delivery line (18) from the elbow (19).

NOTE: Make sure that both the ends of the delivery line are removed before the FCU removal from the engine.

(c) Fuel bypass line (20) from the fluid connector (24).

NOTE: Make sure that both the ends of the bypass line are removed before the FCU removal from the engine.

(d) Air pressure line (14) from metered elbow (13).

(e) Air pressure line (1) from elbow (2).

(3) Disconnect reversing linkage control rods (9 and 15) (Ref. 76-10-00).

(4) Remove four self-locking nuts and washers securing fuel control unit (6) to studs on fuel pump. Carefully remove unit from pump.

(5) Remove coupling (7).

(6) Remove preformed packing (8) from bypass port in pump.

(7) Cap all lines and open ports.

(8) If fuel control unit is being replaced, remove fuel control arm extension (30), fuel control arm (29) and serrated spacer (28) and retain for installation on replacement unit. Install spacer (31), castellated nut (32) and retain with cotterpin (33) on removed unit.

(9) Remove line fittings (2, 13 and 24) and associated parts from respective parts on FCU, and retain for installation on replacement unit.

NOTE: Do not remove elbow fittings (17 and 19).

D. Installation of Fuel Control Unit (PT6A-27 and PT6A-28 Engines) (Ref. Fig. 202)

CAUTION: VARIOUS COMPONENTS ON THE FUEL CONTROL UNIT ARE LOCKWIRED AND SEALED. ENSURE ALL SUCH SEALS ARE INTACT PRIOR TO INSTALLATION AND ARE NOT TAMPERED WITH. UNITS WITH BROKEN SEALS MUST BE RETURNED TO OVERHAUL FOR RECALIBRATION.

NOTE: When a replacement FCU is being installed, check the length of storage and do reconditioning after storage (Ref. Storage).

(1) If fuel control unit to be installed is a replacement unit, depreserve and assemble fittings removed in Para. C.:

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: BEFORE FLUSHING, ENSURE AIR SECTION OF UNIT IS WELL SEALED TO PREVENT CONTAMINATION BY ENTRY OF FUEL.

(a) Remove shipping caps and covers from replacement unit.

(b) Flush fuel section of unit with fuel (PWC01-001), that has been passed through a 10-micron (nominal) filter, until all preservation fluid has been removed. Make sure residual flushing fuel is drained from unit.

NOTE: After installation of elbow (2) and metered elbow (13), do not final torque locknuts (3) and (12). Final torquing will be done at a later stage of installation.

(c) Install elbow (2) with locknut (3), back-up ring (4) and preformed packing (5) (Ref. 70-00-00). Cap elbow to prevent ingress of foreign material until unit is ready for installation.

(d) Install metered elbow (13) with locknut (12), back-up ring (11) and preformed packing (10) (Ref. 70-00-00). Cap elbow to prevent ingress of foreign material until unit is ready for installation.

NOTE: Elbow (13) is unmetered on Pre-SB1123 engines.

(e) Install fluid passage bolt (26) with preformed packing (27). Tighten bolt, torque to 90 to 100 lb.in., and lockwire.

(f) Install preformed packing (25) and fluid connector (24) on passage bolt (26). Secure with locknut (21), back-up ring (22) and preformed packing (23). Locknut to be fingertight.

(2) Install preformed packing (8) in bypass port in mounting face of pump; lubricate with engine oil (PWC03-001).

CAUTION: IF THE FCU SPLINE AND THE PLASTIC DRIVE COUPLING (7) ARE NOT FULLY ENGAGED, IT COULD PUSH AND DISENGAGE THE FUEL PUMP DRIVE SHAFT WHEN MATED WITH THE FUEL PUMP ASSEMBLY.

(3) Install and push the plastic drive coupling (7) on the FCU shaft to make sure that the coupling is correctly engaged on the FCU shaft such that there is no more visible spline is found.

(4) Make sure that the fuel control unit driveshaft bearing retainer plate and surround area is clean. Remove all traces of blue grease or blue dye residue.

CAUTION: MAKE SURE THAT THE EACH STUD PROTRUDES A MINIMUM OF 0.054 IN. FROM EACH NUT.

CAUTION: CARE MUST BE TAKEN TO ALIGN THE SPLINES. DO NOT FORCE THE SPLINES TO ENGAGE THE FCU AND FUEL PUMP AS THIS COULD DISENGAGE THE PUMP SHAFT.

(5) Put the fuel control unit (6) on the studs of the fuel pump. Push in to engage the splines of the coupling (7) with the splines of the pump driveshaft. As the mounting faces meet, make sure that the preformed packing (8) is not displaced. Attach the fuel control unit (6) to the fuel pump with the washers and the nuts. Torque the nuts 75 to 85 lbf.in.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(6) Reconnect fuel and air pressure lines:

(a) Connect fuel bypass line (20) to fluid connector (24); tighten locknut (21), torque to 38 to 42 lb.in., and lockwire. Tighten coupling nut of line, torque to 90 to 100 lb.in., and lockwire.

(b) Connect fuel delivery line (18) to elbow (19). Tighten coupling nut, torque to 90 to 100 lb.in., and lockwire.

NOTE: Do not lockwire connections of air pressure lines. Lockwiring must be done after pressure test (Ref. 73-10-07, Adjustment/Test)

(c) Connect air pressure line (1) to elbow (2); final torque locknut (3) to 38 to 42 lb.in. Tighten coupling nut of line and torque to 90 to 100 lb.in.

(d) Connect air pressure line (14) to elbow and metering plug (13); final torque locknut (12) to 38 to 42 lb.in. Tighten coupling nut of line and torque to 90 to 100 lb.in.

(e) Connect fuel inlet line (16) to elbow (17). If a new unit is installed, torque locknut on elbow to 70 to 80 lb.in. Tighten coupling nut of line, torque to 270 to 300 lb.in., and lockwire.

(f) If new or replacement FCU is being installed, remove cotterpin (33), castellated nut (32) and spacer (31) from FCU and install serrated spacer (28), fuel control arm (29) and FCU arm extension on FCU. Tighten extension fingertight only.

(g) Reconnect FCU reversing interconnect rod (9) and starting control rod (15) to FCU and adjust as required (Ref. 76-10-00).

(h) Lockwire all connection locknuts.

6. Cleaning/Painting

A. General Cleaning

(1) Clean the exterior surfaces of the fuel control unit with solvent (PWC11-027) or (PWC11-031). To prevent the cleaning agent from entering the unit, do not remove the blanking caps and plugs.

(2) Wipe the surfaces dry with clean, lint-free cloths to remove all residual solvent and contamination.

B. Compressor Delivery Px Metering Plug (Ref. Figs. 201 and 202)

CAUTION: AVOID MIXING THE ELBOW (2) AND THE METERED ELBOW (11, Fig. 201; 13, Fig. 202) WITH SIMILAR PARTS AND MAKE SURE THAT THE ORIGINAL METERED ELBOW WITH THE METERING PLUG IS REINSTALLED. DO NOT REMOVE THE METERING PLUG FROM THE ELBOW FOR CLEANING.

(1) If the fittings were not removed from the fuel control unit (11 or 13 as applicable, note the angular position of the metered elbow (11, Fig. 201; 13, Fig. 202), loosen the locknut, and remove the elbow from the unit.

NOTE: Elbow (13) is unmettered on Pre-SB1123 engines.

(2) Flush the elbow and metering plug assembly in solvent (PWC11-027) or (PWC11-031) and

make sure that all contamination is removed.

(3) Clean all foreign matter from the metering plug using an unpainted wooden dowel and flush with solvent (PWC11-027) or (PWC11-031).

(4) Dry the parts with clean, dry compressed air.

(5) Reinstall the elbows on the fuel control unit (1), positioning the elbows at the same angles noted during removal and tighten the locknuts. Torque the locknuts 38 to 42 lb.in., and secure the nut to the FCU with lockwire.

7. Inspection/Check

A. General

(1) Examine mounting flange, housing and components of lever mechanism for cracks and other surface defects. Cracks are not permissible; minor defects may be repaired.

(2) Examine ports for damage to internal threads and, if evident, send unit for rectification at an overhaul facility.

(3) Examine elbow fittings for damage to threads.

(4) Examine external surfaces for corrosion and for general condition of anodic treatment.

(5) Examine the FCU for possible driveshaft bearing grease washout by too much fuel or oil as follows:

NOTE: 1. Too much of fuel or oil can possibly dissolve the driveshaft bearing grease. If you think that grease dissolved, send the FCU to an approved overhaul facility for inspection/repair.

NOTE: 2. FCU bearing grease washout can be an indication of fuel pump rear-side carbon seal leakage, examine the fuel pump for such condition (Ref. 73-10-00, FUEL PUMP - MAINTENANCE PRACTICES).

(a) Do a visual inspection of the driveshaft bearing face and the retaining plate for signs of blue grease. Signs of blue grease permitted. Grease with a hard or jelly-like consistency shows that the area is not contaminated by too much fuel or oil and still lubricates the driveshaft bearing.

(b) Examine the vent hole for signs of blue stain. If there is a stain, then driveshaft bearing grease can possibly be washed out by fuel.

(c) Make sure that the driveshaft turns freely and smoothly when you turn with a light finger force. When you release the finger force, the driveshaft must stop turning.

NOTE: A free turn of driveshaft can be an indication of grease washout and this is the cause for rejection.

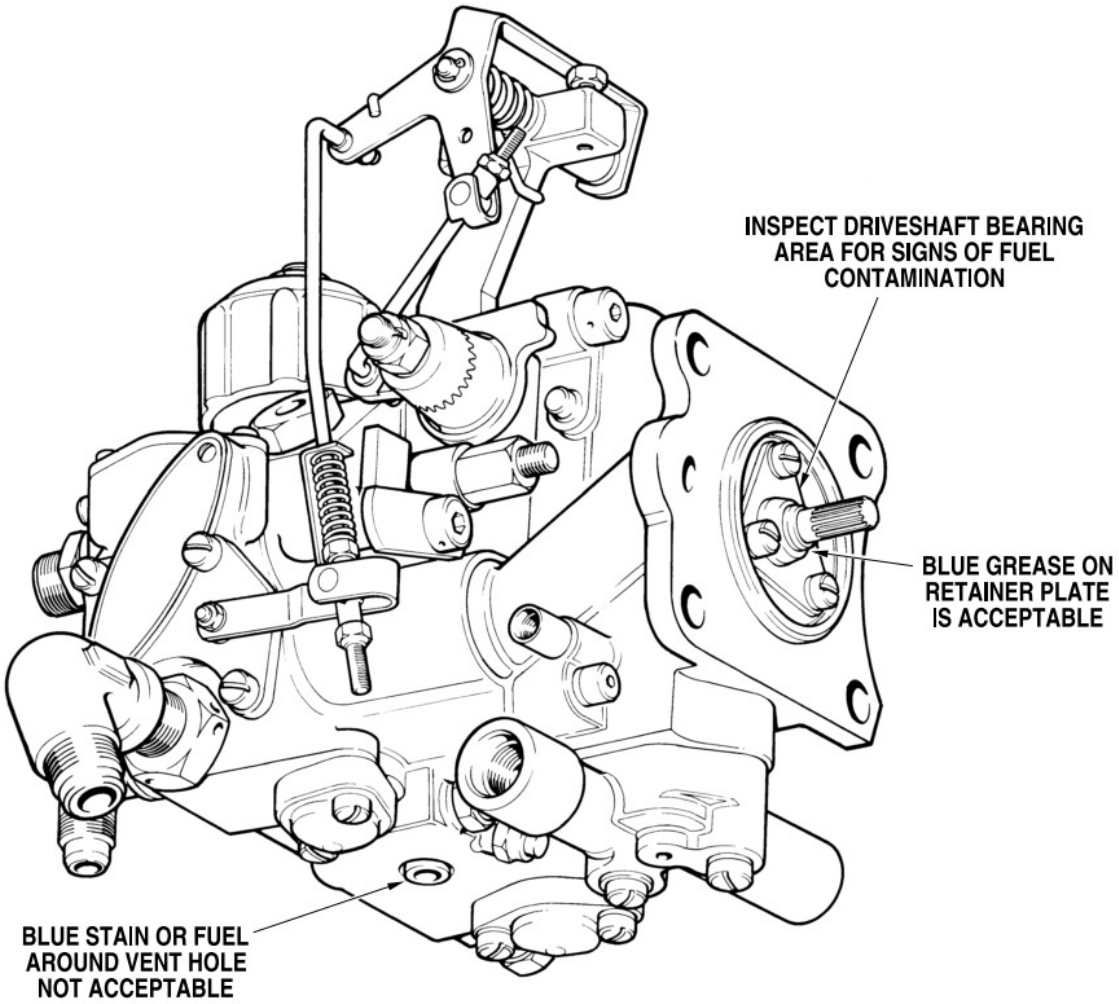
(6) Check the drive couplings and FCU driveshaft for worn splines. Reject unit if any wear is found.

(7) Brinelling or rough operating condition of the driveshaft bearing is the cause for rejection.

B. Leakage Inspection (Ref. Fig. 203)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 203 Fuel Control Unit (FCU) (Typical) - Inspection/Check



c41457a

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

NOTE: Limits are during engine operation.

(1) No leakage is permitted from the fuel pump/FCU casting or the interconnect line.

(2) Check for fuel leaks at the bypass port. If leakage is evident, replace the preformed packing.

(3) Leakage of engine oil or fuel from the fuel pump seal drain port must be assessed in accordance with Chapter 73-10-02, Fuel Pump - Maintenance Practices, Inspection/Check.

(4) Check for evidence of residual blue dye or fuel leaks from the vent hole on the underside of the FCU. Leakage of fuel or blue dye is not acceptable. If leaks are found, reject FCU and send unit for rectification to an approved overhaul facility.

NOTE: Blue dye leaking from the fuel pump seal drain port indicates possible contamination of, or damage to, the FCU driveshaft bearing by grease/lubricant wash-out.

(5) Do an assessment of the driveshaft bearing serviceability in accordance with paragraphs A.(5) to A.(7). If bearing wash-out is suspected, reject the FCU.

(6) Check for signs of external leakage.

(7) Examine external surfaces for corrosion and for general condition of the anodic treatment.

C. Lockwire and Seals

(1) Examine the lockwire and seals for security and damage. If any seal indicates tampering, or if the lockwire is broken, ship the unit to an overhaul facility for recalibration.

CAUTION: TO MAKE SURE CALIBRATION AND OPERATION HAS NOT BEEN AFFECTED, DO NOT REPLACE A LOCKWIRED SEAL UNTIL THE UNIT HAS BEEN CHECKED.

(2) Examine the remainder of the lockwire and replace if damaged or broken.

8. Approved Repairs

A. General Repairs

NOTE: Clean all areas and apply corrosion preventive treatment to aluminum surfaces after carrying out minor repairs. Do not allow filings and dirt to enter the fuel control unit during the repair.

(1) Minor surface damage such as burrs, nicks, scores, scratches and similar defects may be cleaned up by local blending with a fine stone or crocus cloth (PWC05-061), making sure that all high spots and sharp edges are removed.

(2) Clean up minor thread damage in the drain and Po ports with a suitable swiss file, or chase threads 7/16-20 UNJF-3B.

(3) Clean up minor thread damage on the elbow fittings with a suitable swiss file, or chase with appropriately sized die. Replace the fittings if thread damage is severe.

B. Corrosion Removal

NOTE: Only light surface corrosion may be removed from the housing. Severe corrosion,

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

indicated by surface etching and/or heavy pitting, is cause for rejection of the unit.

(1) Remove the corrosion using one of the following methods:

(a) Mask off the areas around the corrosion and vapor blast to remove the corrosion, using No. 1200 grit (PWC05-168) or finer.

(b) Carefully polish with No. 400 grit abrasive cloth (PWC05-101) followed by crocus cloth (PWC05-061) and blend into the surrounding area.

(2) Apply anti-corrosion treatment, in accordance with the procedure given in Subpara. C. following, to all repair areas.

C. Application of Anodize Surface Treatment

(1) The anodic anti-corrosion finish on the surfaces of the housings may be repaired by local application of a chemical treatment solution (PWC05-166). This treatment must be used on areas where minor repairs have been done by local blending and/or polishing. To apply the treatment, proceed as follows:

(a) Prepare the surface by swabbing with cleaning solution (PWC11-038). Keep the surface wetted with cleaning agent for one to five minutes.

(b) Rinse with clean water to remove all traces of the cleaning agent.

WARNING: ALODINE SOLUTION IS EXTREMELY DANGEROUS. IT CONTAINS AN OXIDIZING INGREDIENT WHICH CAN CAUSE AN EXPLOSION IF IT COMES IN CONTACT WITH COMBUSTIBLE MATERIALS SUCH AS PAINTS AND SOLVENTS.

WARNING: DURING THE USE OF ALODINE SOLUTION, AVOID BREATHING THE VAPORS; USE ONLY WITH ADEQUATE VENTILATION. WEAR CHEMICAL SAFETY GOGGLES AND RUBBER GLOVES WHEN HANDLING. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER; IF IRRITATION APPEARS AND PERSISTS, CONSULT A PHYSICIAN (MEDICAL DOCTOR). IN CASE OF EYE CONTACT, FLUSH EXTENSIVELY WITH FRESH WATER AND CONSULT A PHYSICIAN IMMEDIATELY.

(c) Apply the chemical treatment solution (PWC05-166) to the prepared surface by brush, swab, or spray method and allow three to five minutes setting time.

(d) Rinse with clean water to remove all excess chemical treatment solution and allow the surface to air dry.

(e) Examine the coating and make sure the repair area is completely covered. Reapply the treatment as necessary.

9. Adjustment/Test

Do a functional check of the fuel control unit during the next engine test run (Ref. 71-00-00).

10. Storage

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

A. Preservation of FCU

(1) Short term storage - 28 days or less:

CAUTION: DO NOT PERMIT FUEL OR OIL TO ENTER THE DRIVE BODY CAVITY OR ANY AIR PRESSURE PORTS.

(a) Fill the fuel section of the FCU with calibrating fluid (PWC03-002) filtered through a 10 micron filter.

(b) Install caps and plugs in all ports of the fuel section of the FCU.

(c) Check the security of all caps and plugs in all ports of the air section of the FCU to prevent the entrance of dirt or other contaminates.

(d) Put a tag on the unit indicating date of storage.

(e) Refer to Para. C., Packaging and Shipping, for protection from dirt and other contaminants.

(f) Every ten days, check the fluid level in the fuel section of the unit and refill as necessary.

(2) Long term storage - more than 28 days:

(a) Do an inspection of the FCU for general condition and correct driveshaft grease condition (Ref. Para. 7.A.).

CAUTION: DO NOT PERMIT FUEL OR OIL TO ENTER THE DRIVE BODY CAVITY OR ANY AIR PRESSURE PORTS.

(b) Drain all remaining fluid from the FCU.

(c) Fill the fuel section of the FCU with preservative oil (PWC05-077) filtered through a 10 micron filter.

NOTE: Tip the FCU as necessary to make sure that a complete film of oil goes into all ports and passages of the fuel section of the unit.

(d) Drain the excess oil as required and install caps and plugs in all ports of the fuel section of the unit.

(e) Check the security of all caps and plugs in all openings of the air section of the FCU to prevent the entrance of dirt or other contaminates.

(f) Put a tag on the unit indicating date of storage.

(g) Refer to Para. C., Packaging and Shipping, for protection from dirt and other contaminants.

B. Depreservation of FCU - Reconditioning after Storage

CAUTION: DO NOT PERMIT FUEL OR OIL TO ENTER THE DRIVE BODY CAVITY OR ANY AIR PRESSURE PORTS.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: DO NOT ATTEMPT TO REPLACE ANY LOCKWIRE SEAL UNTIL THE FCU HAS BEEN CHECKED TO BE CERTAIN THAT CALIBRATION OR OPERATION HAS NOT BEEN EFFECTED.

(1) Do an inspection of the FCU for general condition and correct driveshaft grease condition (Ref. Para. 7.A.).

NOTE: The storage life limit of the Honeywell Fuel Control Unit (FCU) is six years (Ref. 72-00-00, ENGINE TURBOPROP - SERVICING). If you stored the FCU for more than six years, send the FCU for overhaul.

(2) Inspect all lockwire and lockwire seals. Replace broken lockwire after checking the torque value of nuts, plugs, caps or screws that were lockwired.

(3) Return the unit for recalibration if any lockwire seal indicates tampering or if the lockwire is broken.

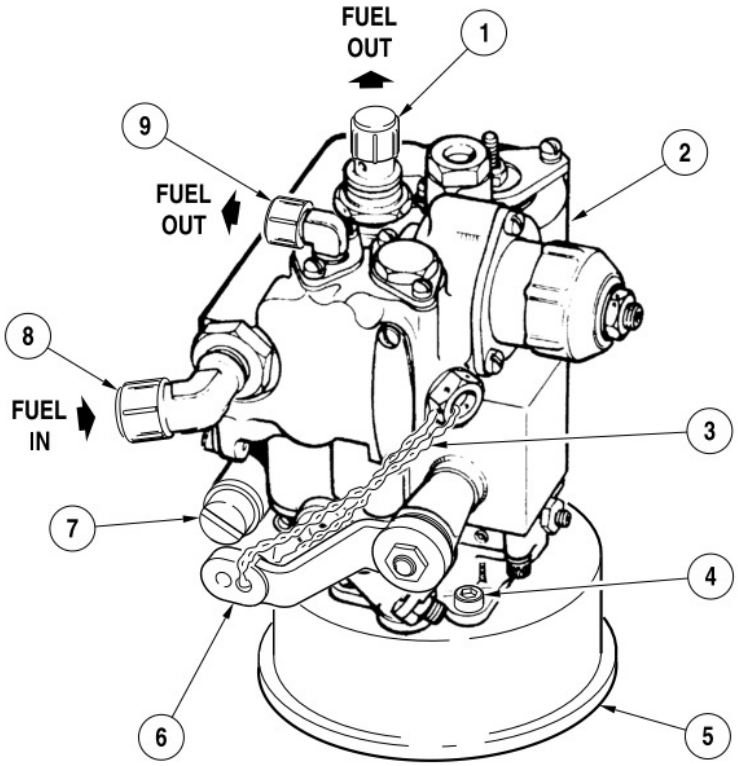
(4) Fuel control units may be shipped to:

Pratt & Whitney Canada Corp.
Plant 5, Accessories Business
7007, De La Savane
St-Hubert, QC
Canada
J3Y 3X7

C. Packaging and Shipping of FCU (Ref. Fig. 204)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 204 Fuel Control Unit (Typical) - Storage



c88316

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Fuel Bypass Connector and Shipping Cap/Plug
2. Fuel Control Unit
3. Lockwire
4. Bolt
5. Shipping Stand (P/N 2529654)
6. Power Lever
7. P3 Air Port and Shipping Cap/Plug
8. Fuel Inlet Connector and Shipping Cap/Plug
9. Fuel Pressure Connector and Shipping Cap/Plug

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(1) When the FCU is to be shipped or stored, prepare the unit as follows:

(a) Make sure that all shipping plugs and caps are secure.

(b) Place the FCU (2) on the shipping base (5) and secure with bolts (4).

CAUTION: THE POWER LEVER MUST BE SECURED DURING STORAGE OR SHIPPING OR ELSE DAMAGE TO THE DRIVE SHAFT OR POWER LEVER CAN RESULT.

(c) Put the power lever (6) in position and secure with lockwire (3) as shown to prevent movement during storage or shipping.

(d) Put the FCU and stand into a moisture and vapor proof container or plastic bag and seal the opening. Attach a tag to the unit indicating the date of storage.

(e) Store the unit in an approved shipping carton or case.

(2) The shipping base P/N 2529654 can be purchased from:

Air Parts and Supply Company
12840 SW 84th. Avenue Rd.,
Miami, FL 33156
USA
TEL: (305) 235-5401
FAX: (305) 235-8185

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement